

Genes specify proteins via transcription and translation

Inherited traits like albinism are determined by genes, which have information content in the form of specific sequences of nucleotides along stretches of DNA, the genetic material. The DNA inherited by an organism leads to specific traits by dictating the synthesis of proteins and of RNA molecules involved in protein synthesis. In the case of coat and skin color, the gene containing the information to synthesize an enzyme that makes pigment allows normal coloring, and a faulty version of that gene leads to lack of color, or the albino phenotype. Proteins are the link between genotype and phenotype. **Gene expression** is the process by which DNA directs the synthesis of proteins (or, in some cases, just RNAs). The expression of genes that code for proteins includes two stages: transcription and translation.

Before going into the details of how genes direct protein synthesis, let's step back and examine how the fundamental relationship between genes and proteins was discovered.

Evidence from Studying Metabolic Defects

In 1902, British physician Archibald Garrod was the first to suggest that genes dictate phenotypes through enzymes, proteins that catalyze specific chemical reactions in the cell. He postulated that the symptoms of an inherited disease reflect an inability to make a particular enzyme. He later referred to such diseases as “inborn errors of metabolism.” For example, people with a disease called alkaptonuria have black urine because it contains a chemical called alkapton, which darkens upon exposure to air. Garrod reasoned that these people cannot make an enzyme that breaks down alkapton, so alkapton is expelled in their urine.

Several decades later, research supported Garrod's hypothesis that a gene dictates the production of a specific enzyme, later named the *one gene–one enzyme hypothesis*. Biochemists learned that cells synthesize and degrade most organic molecules via metabolic pathways, in which each chemical reaction in a sequence is catalyzed by a specific enzyme (see Concept 8.1). Such metabolic pathways lead, for instance, to the synthesis of the pigments that give the brown donkey in **Figure 17.1** its fur color or fruit flies (*Drosophila*) their eye color (see Figure 15.3). In the 1930s, the American biochemist and geneticist George Beadle and his French colleague Boris Ephrussi speculated that in *Drosophila*, each mutation affecting eye color blocks pigment synthesis at a specific step by preventing production of the enzyme that catalyzes that step. But neither the chemical reactions nor the enzymes that catalyze them were known at the time.

Nutritional Mutants in *Neurospora*: Scientific Inquiry

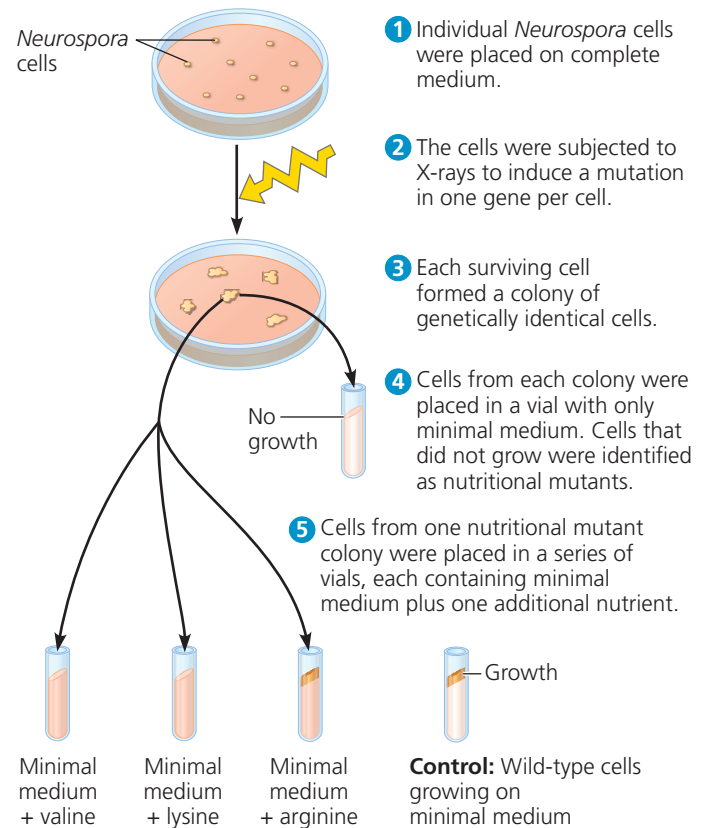
A breakthrough in demonstrating the relationship between genes and enzymes came a few years later at Stanford University, where Beadle and Edward Tatum began working with a bread mold, *Neurospora crassa*. Unlike the diploid

organisms studied by Mendel (peas) and T. H. Morgan (fruit flies), *Neurospora* is a haploid species. Because its genome contains only one copy of each gene, that single copy determines the individual's expressed phenotype. Therefore, when Beadle and Tatum wanted to discover the function of any gene, they only needed to mutate and disable that one allele. They chose to study the protein-coding genes required for a specific nutritional activity. They caused mutations in genes by bombarding *Neurospora* with X-rays and looked among the survivors for mutants that differed in their nutritional needs from the wild-type bread mold.

Wild-type *Neurospora* has modest food requirements. It can grow in the laboratory on *minimal medium*—a simple solution containing minimal nutrients for growth—inorganic salts, glucose, and the vitamin biotin—incorporated into agar, a support medium. From these basics, wild-type mold cells use their metabolic pathways to produce all the other nutrient molecules (like amino acids) they need for growth, dividing repeatedly and forming visible colonies of genetically identical cells. As shown in **Figure 17.2**, Beadle and Tatum generated different “nutritional mutants” of *Neurospora* cells, each of which had a mutation in one gene and was unable to synthesize a particular

▼ Figure 17.2 Beadle and Tatum's experimental approach.

To obtain nutritional mutants, Beadle and Tatum exposed *Neurospora* cells to X-rays, inducing mutations, then screened for (tested for) mutants that had new nutritional requirements, such as arginine, as shown here.



6 The vials were observed for growth. In this example, the mutant cells grew only on minimal medium + arginine, indicating that this mutant was missing the enzyme for the synthesis of arginine.